

The Beveridge Response to a Destruction Shock under Infinitely Elastic Vacancy Creation

Notes for review (not for the draft), October 7, 2026

Numbers are produced by Programs `baseline/gs_free_entry_beveridge.jl` (output `gs_free_entry_bev` block “Exposition tables”). The program checks every closed form below by assertion against a simulated impulse response of the same linear system. Equation numbers “GS (*n*)” refer to Gabrovski and Silva (2025), *Journal of Economic Dynamics and Control* 181, 105205.

1. Setting

We study the Gabrovski–Silva (GS) economy with one change: the destruction rate δ_t is a time-varying shock. In the draft this is the AGS economy of `prop:ags` with $\sigma = \kappa = 0$, which `rem:nesting` identifies as GS extended with aggregate shocks to δ_t and s_t . Its labor-market block closes on its own, as under the conditions of `prop:independence`. The discount rate is r , so the discount factor is $1/(1+r)$. Output per match, y (recruiter compensation z/μ in our model), is constant. Exit is exogenous, so a business survives from t to $t+1$ with probability $1 - \delta_t$. Every vacated position is reposted at no cost, and there are no flow recruiting costs, so all recruiting costs are sunk.

Shock process.

$$\delta_t = \bar{\delta} \exp(\tilde{\delta}_t), \quad \tilde{\delta}_t = \rho \tilde{\delta}_{t-1} + \varepsilon_t, \quad \rho \in [0, 1),$$

with ε_t mean zero, so $\tilde{\delta}_t$ is a mean-zero AR(1) and is the log deviation of δ_t . The separation probability s is constant, so the total separation rate

$$\tau_t \equiv 1 - (1 - \delta_t)(1 - s)$$

moves with δ_t . Its steady state is $\bar{\tau} = 1 - (1 - \bar{\delta})(1 - s)$.

Timing. We follow GS (their Fig. 2) and the draft (S10). δ_t is known at the start of t . Entry, production and matching take place within t . Separation and destruction take place at the end of t , at rates s and δ_t , and a match formed at t is exposed to destruction at the end of t but

not to separation. So u_t and the vacancies inherited from $t - 1$ are fixed at t , while tightness θ_t and entry e_t respond at t .

Matching. $q(\theta) = A\theta^{-\eta_L}$ and $f(\theta) = \theta q(\theta)$, with $\eta_L \in (0, 1)$. GS use the den Haan–Ramey–Watson function; we use Cobb–Douglas, as in our calibration.

2. The nonlinear system

2.1. Vacancy value and flow value

A business opportunity costs x , drawn from $G(x) = (x/x_m)^\xi$ on $[0, x_m]$, and is taken if and only if $x \leq Q_t$. So $e_t = G(Q_t)$ and $Q_t = e_t^{1/\xi} x_m$. Under infinitely elastic creation ($\xi \rightarrow \infty$),

$$Q_t = x_m \quad \text{for all } t. \quad (1)$$

The flow value of a vacancy is defined as in GS (2), with the survival factor dated t :

$$K_t \equiv Q_t - \frac{1 - \delta_t}{1 + r} \mathbb{E}_t Q_{t+1} = x_m \frac{r + \delta_t}{1 + r}. \quad (2)$$

Whereas Q is constant, K_t is a known function of the exogenous δ_t . In GS δ is constant, so free entry fixes K as well (GS (13)). Here δ_t is the depreciation rate of the vacancy asset, so its flow value moves with δ_t . Holding K fixed would give a flow-cost DMP model (Remark 2). We substitute (2) wherever K appears.

2.2. Wage and job creation

The Nash wage is GS (7):

$$w_t = \phi[y - K_t + \theta_t K_t] + (1 - \phi)b. \quad (3)$$

The job creation condition is GS (10), with the survival factor dated t :

$$\frac{K_t}{q(\theta_t)} = \frac{1 - \delta_t}{1 + r} \mathbb{E}_t \left[y - w_{t+1} - K_{t+1} + (1 - s) \frac{K_{t+1}}{q(\theta_{t+1})} \right]. \quad (4)$$

The left side is the expected cost of a hire: the flow value of a vacancy times the expected time to fill. The right side is the expected profit from a match, which first produces at $t + 1$, and only if the business survives destruction at the end of t . Profit has two parts: the flow profit net of the forgone vacancy value, $y - w_{t+1} - K_{t+1}$, and the hiring cost saved if the match continues, $(1 - s)K_{t+1}/q(\theta_{t+1})$.

Using the Nash wage (3) dated $t + 1$,

$$y - w_{t+1} - K_{t+1} = (1 - \phi)(y - b) - K_{t+1}[(1 - \phi) + \phi\theta_{t+1}].$$

So the term in brackets in (4) is

$$(1 - \phi)(y - b) + K_{t+1}\left[\frac{1 - s}{q(\theta_{t+1})} - \phi\theta_{t+1} - (1 - \phi)\right].$$

Multiply (4) by $1 + r$ and substitute (2) at t and $t + 1$. The job creation condition becomes an equation in θ and δ alone:

$$\frac{x_m(r + \delta_t)}{q(\theta_t)} = (1 - \delta_t) \mathbb{E}_t \left\{ (1 - \phi)(y - b) + \frac{x_m(r + \delta_{t+1})}{1 + r} \left[\frac{1 - s}{q(\theta_{t+1})} - \phi\theta_{t+1} - (1 - \phi) \right] \right\}. \quad (5)$$

The left-hand side represents the average hiring cost. The right-hand side represents the benefit of a filled job. Its first term, $(1 - \phi)(y - b)$, is the firm's share of the match's flow surplus. In the second term, $x_m(r + \delta_{t+1})/(1 + r) = K_{t+1}$ is next period's flow value of a vacancy. It is multiplied by $(1 - s)/q(\theta_{t+1})$, the hiring cost saved if the match continues; by $-\phi\theta_{t+1}$, the worker's share of that saving, $\phi fK/q$; and by $-(1 - \phi)$, the firm's share of the forgone flow value of the vacancy.

2.3. Laws of motion

Vacancies follow GS (8). Unemployment follows GS (9), with δ and τ dated $t - 1$ because the destruction they cause takes place at the end of $t - 1$:

$$v_t = (1 - \delta_{t-1})[1 - q(\theta_{t-1})]v_{t-1} + (1 - \delta_{t-1})s(1 - u_{t-1}) + e_t, \quad (6)$$

$$u_t = [1 - (1 - \delta_{t-1})f(\theta_{t-1})]u_{t-1} + \tau_{t-1}(1 - u_{t-1}), \quad (7)$$

$$\theta_t = v_t/u_t. \quad (8)$$

The vacancy pool (6) has three parts: unmatched vacancies from $t - 1$ that avoided destruction, vacancies reposted after a separation at a surviving business, and new entrants e_t . The first two are fixed before t . Call their sum the predetermined stock:

$$v_{pre,t} \equiv (1 - \delta_{t-1})\left\{ [1 - q(\theta_{t-1})]v_{t-1} + s(1 - u_{t-1}) \right\}, \quad v_t = v_{pre,t} + e_t.$$

Every variable on its right side is dated $t - 1$, which is what makes it predetermined at t . Unemployment (7) has two parts: last period's unemployed who were not hired, or were hired at a business that was then destroyed, and last period's employed who separated, at rate τ_{t-1} .

How free entry orders the system. Equation (5) contains only θ and the exogenous δ , so it determines the path of θ on its own. Equation (7) then gives u , (8) gives $v = \theta u$, and (6) determines entry, $e_t = v_t - v_{pre,t}$. With finitely elastic creation the causation reverses: entry is set by Q , (6) determines v , and (4) determines Q .

3. Steady state

Following GS's Definition 2, with $Q = x_m$, the steady state satisfies

$$\frac{\bar{K}}{q(\bar{\theta})} = \frac{1 - \bar{\delta}}{r + \bar{\tau}}(y - \bar{w} - \bar{K}), \quad (9)$$

$$\bar{K} = x_m \frac{r + \bar{\delta}}{1 + r}, \quad (10)$$

$$\bar{w} = \phi[y - \bar{K} + \bar{\theta}\bar{K}] + (1 - \phi)b, \quad (11)$$

$$\bar{u} = \frac{\bar{\tau}}{\bar{\tau} + (1 - \bar{\delta})f(\bar{\theta})}, \quad (12)$$

$$\bar{e} = \bar{\delta}(\bar{v} + 1 - \bar{u}), \quad (13)$$

$$\bar{\theta} = \bar{v}/\bar{u}.$$

These are GS (12)–(16).

Equation (9) is the steady state of (4),

$$\frac{\bar{K}}{\bar{q}} = \frac{1 - \bar{\delta}}{1 + r} \left[y - \bar{w} - \bar{K} + (1 - s) \frac{\bar{K}}{\bar{q}} \right],$$

multiplied by $1 + r$ and rearranged using $(1 + r) - (1 - \bar{\delta})(1 - s) = r + \bar{\tau}$. Equation (10) is a user cost: the asset price x_m times the discount rate plus the destruction rate. The share of the user cost due to destruction is $\bar{\delta}/(r + \bar{\delta})$. Equation (12) can be written $(1 - \bar{\delta})\bar{f}\bar{u} = \bar{\tau}(1 - \bar{u})$: hires equal separations. Since $\bar{f}\bar{u} = \bar{q}\bar{v}$, this is also $(1 - \bar{\delta})\bar{q}\bar{v} = \bar{\tau}(1 - \bar{u})$.

For (13), the steady state of (6) gives

$$\bar{e} = \bar{v} - (1 - \bar{\delta})(1 - \bar{q})\bar{v} - (1 - \bar{\delta})s(1 - \bar{u}) = \bar{\delta}\bar{v} + (1 - \bar{\delta})\bar{q}\bar{v} - (1 - \bar{\delta})s(1 - \bar{u}).$$

Using hires equal separations, this is $\bar{\delta}\bar{v} + [\bar{\tau} - (1 - \bar{\delta})s](1 - \bar{u})$, and $\bar{\tau} - (1 - \bar{\delta})s = \bar{\delta}$. So entry replaces the positions destroyed, both vacant ($\bar{v}$) and filled ($1 - \bar{u}$). Positions vacated by separations are reposted and need no replacement. The shares of entrants and of predetermined vacancies in the stock are therefore

$$\frac{\bar{e}}{\bar{v}} = \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}}, \quad \frac{\bar{v}_{pre}}{\bar{v}} = 1 - \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}}.$$

Filled positions far outnumber vacancies, $(1 - \bar{u})/\bar{v} \approx 26$ at our targets, so the entrant share is small when destruction is rare. Hires equal separations gives $(1 - \bar{u})/\bar{v} = (1 - \bar{\delta})\bar{q}/\bar{\tau}$: filled jobs per vacancy equal hires per vacancy times the expected duration of a job. An equivalent form of the entrant share is therefore

$$\frac{\bar{e}}{\bar{v}} = \frac{\bar{\delta}}{\bar{\tau}} [\bar{\tau} + (1 - \bar{\delta})\bar{q}].$$

Our calibration fixes $\bar{\tau}$ and the gross vacancy-filling rate $(1 - \bar{\delta})\bar{q}$, so the bracket is the same, 0.831, in every economy we compare. The entrant share is then proportional to the destruction share of separations, $\bar{\delta}/\bar{\tau}$.

4. Log-linear system

Hats denote log deviations from the steady state, $\hat{x}_t \equiv \ln x_t - \ln \bar{x}$. The shock is already in logs: $\hat{\delta}_t = \tilde{\delta}_t$, so $\delta_t - \bar{\delta} \approx \bar{\delta} \tilde{\delta}_t$. The separation rate moves by $\tau_t - \bar{\tau} = (1 - s)(\delta_t - \bar{\delta})$, so

$$\hat{\tau}_t = (1 - s) \frac{\bar{\delta}}{\bar{\tau}} \tilde{\delta}_t.$$

A one-percent rise in δ raises τ by $(1 - s)\bar{\delta}/\bar{\tau}$ percent, the destruction share of separations up to the factor $1 - s$.

4.1. Job creation

Take logs of both sides of (5). On the left,

$$\ln \frac{x_m(r + \delta_t)}{q(\theta_t)} = \ln x_m + \ln(r + \bar{\delta}e^{\tilde{\delta}_t}) - \ln A + \eta_L \ln \theta_t,$$

whose first-order deviation is

$$\frac{\bar{\delta}}{r + \bar{\delta}} \tilde{\delta}_t + \eta_L \hat{\theta}_t.$$

On the right, the log of $1 - \delta_t$ deviates by $-[\bar{\delta}/(1 - \bar{\delta})]\tilde{\delta}_t$. Write Σ_{t+1} for the term in braces in (5). In the steady state of (5), $x_m(r + \bar{\delta})/\bar{q} = (1 - \bar{\delta})\bar{\Sigma}$. Only the second term of Σ_{t+1} varies, and its deviation is

$$d\Sigma_{t+1} = \frac{x_m}{1 + r} \left\{ \bar{\delta} \tilde{\delta}_{t+1} \left[\frac{1 - s}{\bar{q}} - \phi\bar{\theta} - (1 - \phi) \right] + (r + \bar{\delta}) \left[(1 - s) \frac{\eta_L}{\bar{q}} - \phi\bar{\theta} \right] \hat{\theta}_{t+1} \right\},$$

using $d(1/q) = (\eta_L/\bar{q})\hat{\theta}$ and $d\theta = \bar{\theta}\hat{\theta}$. Divide by $\bar{\Sigma} = x_m(r + \bar{\delta})/[(1 - \bar{\delta})\bar{q}]$, and use $\bar{\theta}\bar{q} = \bar{f}$:

$$\hat{\Sigma}_{t+1} = \frac{1 - \bar{\delta}}{1 + r} \left\{ [1 - s - \phi\bar{f} - (1 - \phi)\bar{q}] \frac{\bar{\delta}}{r + \bar{\delta}} \tilde{\delta}_{t+1} + [(1 - s)\eta_L - \phi\bar{f}] \hat{\theta}_{t+1} \right\}.$$

Distribute $1 - \bar{\delta}$ inside the braces. Since $(1 - \bar{\delta})(1 - s) = 1 - \bar{\tau}$,

$$\hat{\Sigma}_{t+1} = \frac{1}{1+r} \left\{ [1 - \bar{\tau} - \phi(1 - \bar{\delta})\bar{f} - (1 - \phi)(1 - \bar{\delta})\bar{q}] \frac{\bar{\delta}}{r + \bar{\delta}} \tilde{\delta}_{t+1} + [(1 - \bar{\tau})\eta_L - \phi(1 - \bar{\delta})\bar{f}] \hat{\theta}_{t+1} \right\}.$$

The matching rates now appear as $(1 - \bar{\delta})\bar{f}$ and $(1 - \bar{\delta})\bar{q}$, the effective job-finding and vacancy-filling rates: the probabilities that a match forms and its business survives to production. These are the rates we calibrate, 0.41 and 0.80. Setting the left-side deviation equal to $-\bar{\delta}/(1 - \bar{\delta})\tilde{\delta}_t + \mathbb{E}_t \hat{\Sigma}_{t+1}$, moving the $\tilde{\delta}_t$ term to the right and collecting terms,

$$\boxed{\eta_L \hat{\theta}_t = - \left[\frac{\bar{\delta}}{r + \bar{\delta}} + \frac{\bar{\delta}}{1 - \bar{\delta}} \right] \tilde{\delta}_t + \frac{1}{1+r} \mathbb{E}_t \left\{ [1 - \bar{\tau} - \phi(1 - \bar{\delta})\bar{f} - (1 - \phi)(1 - \bar{\delta})\bar{q}] \frac{\bar{\delta}}{r + \bar{\delta}} \tilde{\delta}_{t+1} + [(1 - \bar{\tau})\eta_L - \phi(1 - \bar{\delta})\bar{f}] \hat{\theta}_{t+1} \right\}} \quad (14)$$

Only $\hat{\theta}$ and the shock appear. Next period's terms are discounted at $1/(1+r)$, and survival enters through the probability that the match continues, $1 - \bar{\tau}$, and through the effective matching rates.

The weights in (14) read as follows. The coefficient η_L on $\hat{\theta}_t$ says that a tighter market lengthens the time to fill, so a hire costs more. The weight $\bar{\delta}/(r + \bar{\delta})$ on $\tilde{\delta}_t$ is the flow-value effect: a one-percent rise in δ raises the flow value $K_t = x_m(r + \delta_t)/(1 + r)$ by $\bar{\delta}/(r + \bar{\delta})$ percent, the share of the user cost due to destruction, and so raises the cost of a hire. This share is close to one when $\bar{\delta} \gg r$ and close to $\bar{\delta}/r$ when $\bar{\delta} \ll r$. The weight $\bar{\delta}/(1 - \bar{\delta})$ is the survival effect: a hire pays off only if the business survives, so destruction lowers the value of a hire. It is about $\bar{\delta}$, which is small.

The terms in $\tilde{\delta}_{t+1}$ are nonzero only if the shock persists. A higher flow value next period raises the hiring cost saved by a match that continues, with probability $1 - \bar{\tau}$. Part of that saving goes to the worker through her outside option, $-\phi(1 - \bar{\delta})\bar{f}$. A higher flow value also lowers the firm's flow profit, since a matched firm forgoes it, $-(1 - \phi)(1 - \bar{\delta})\bar{q}$. In the same way, a tighter market next period raises the hiring cost a continuing match saves, $(1 - \bar{\tau})\eta_L$, but it also raises the worker's outside option and hence the wage, $-\phi(1 - \bar{\delta})\bar{f}$.

4.2. Unemployment

Write the right side of (7) as $U(u_{t-1}, \delta_{t-1}, \tau_{t-1}, \theta_{t-1}) = [1 - (1 - \delta_{t-1})A\theta_{t-1}^{1-\eta_L}]u_{t-1} + \tau_{t-1}(1 - u_{t-1})$. Its partial derivatives at the steady state are

$$\frac{\partial U}{\partial u_{t-1}} = 1 - (1 - \bar{\delta})\bar{f} - \bar{\tau}, \quad \frac{\partial U}{\partial \delta_{t-1}} = \bar{f}\bar{u},$$

$$\frac{\partial U}{\partial \tau_{t-1}} = 1 - \bar{u}, \quad \frac{\partial U}{\partial \theta_{t-1}} = -(1 - \bar{\delta}) \bar{u} (1 - \eta_L) \frac{\bar{f}}{\bar{\theta}}.$$

Hence, with $\delta_{t-1} - \bar{\delta} = \bar{\delta} \tilde{\delta}_{t-1}$ and $\tau_{t-1} - \bar{\tau} = \bar{\tau} \hat{\tau}_{t-1}$,

$$u_t - \bar{u} = [1 - (1 - \bar{\delta}) \bar{f} - \bar{\tau}] (u_{t-1} - \bar{u}) + \bar{f} \bar{u} \bar{\delta} \tilde{\delta}_{t-1} + (1 - \bar{u}) \bar{\tau} \hat{\tau}_{t-1} - (1 - \bar{\delta}) \bar{u} (1 - \eta_L) \bar{f} \hat{\theta}_{t-1}.$$

Divide by $\bar{u}$, and use $(1 - \bar{u}) \bar{\tau} / \bar{u} = (1 - \bar{\delta}) \bar{f}$ from (12). Define

$$\lambda \equiv 1 - (1 - \bar{\delta}) \bar{f} - \bar{\tau}, \quad \omega \equiv (1 - \bar{\delta}) \bar{f} (1 - \eta_L).$$

Then

$$\hat{u}_t = \lambda \hat{u}_{t-1} + \bar{f} \bar{\delta} \tilde{\delta}_{t-1} + (1 - \bar{\delta}) \bar{f} \hat{\tau}_{t-1} - \omega \hat{\theta}_{t-1}. \quad (15)$$

Substitute $\hat{\tau}_{t-1} = (1 - s)(\bar{\delta} / \bar{\tau}) \tilde{\delta}_{t-1}$. The two coefficients on $\tilde{\delta}_{t-1}$ combine as

$$\bar{f} \bar{\delta} + (1 - \bar{\delta})(1 - s) \bar{f} \frac{\bar{\delta}}{\bar{\tau}} = \bar{f} \frac{\bar{\delta}}{\bar{\tau}} [\bar{\tau} + (1 - \bar{\tau})] = \bar{f} \frac{\bar{\delta}}{\bar{\tau}},$$

because $(1 - \bar{\delta})(1 - s) = 1 - \bar{\tau}$. So

$$\boxed{\hat{u}_t = \lambda \hat{u}_{t-1} + \bar{f} \frac{\bar{\delta}}{\bar{\tau}} \tilde{\delta}_{t-1} - \omega \hat{\theta}_{t-1}} \quad (16)$$

The coefficient λ is persistence. An extra unemployed worker is still unemployed next month unless hired at a surviving business, with probability $1 - (1 - \bar{\delta}) \bar{f}$, and the $\bar{\tau}$ term reflects that one fewer employed worker means one fewer separation. The destruction weight has two channels, visible in (15). The first, $\bar{f} \bar{\delta}$, is lost matches: new matches at a destroyed business never produce. The second is separations. In steady state the inflow into unemployment equals the outflow, $(1 - \bar{\delta}) \bar{f} \bar{u}$, so a one-percent rise in separations raises unemployment by $(1 - \bar{\delta}) \bar{f}$ percent, and separations rise by $(1 - s) \bar{\delta} / \bar{\tau}$ percent per percent of δ . Together the two channels give $\bar{f} \bar{\delta} / \bar{\tau}$, and the second dominates. The unemployment loading is therefore proportional to the destruction share of separations, $\bar{\delta} / \bar{\tau}$. Finally, ω is the hiring channel: lower tightness at $t - 1$ means fewer hires, and those workers are still unemployed at t .

4.3. Vacancies

Take the deviation of $v_t = v_{pre,t} + e_t$ and divide by $\bar{v}$:

$$\hat{v}_t = \frac{\bar{v}_{pre}}{\bar{v}} \hat{v}_{pre,t} + \frac{\bar{e}}{\bar{v}} \hat{e}_t.$$

Substituting the shares implied by (13),

$$\hat{v}_t = \left[1 - \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}} \right] \hat{v}_{pre,t} + \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}} \hat{e}_t \quad (17)$$

For the predetermined stock, the partial derivatives of $v_{pre,t} = (1 - \delta_{t-1})\{[1 - A\theta_{t-1}^{-\eta_L}]v_{t-1} + s(1 - u_{t-1})\}$ at the steady state are

$$\begin{aligned} \frac{\partial v_{pre,t}}{\partial \delta_{t-1}} &= -[(1 - \bar{q})\bar{v} + s(1 - \bar{u})] = -\frac{\bar{v}_{pre}}{1 - \bar{\delta}}, & \frac{\partial v_{pre,t}}{\partial v_{t-1}} &= (1 - \bar{\delta})(1 - \bar{q}), \\ \frac{\partial v_{pre,t}}{\partial \theta_{t-1}} &= (1 - \bar{\delta})\eta_L \frac{\bar{q}}{\bar{\theta}} \bar{v}, & \frac{\partial v_{pre,t}}{\partial u_{t-1}} &= -(1 - \bar{\delta})s. \end{aligned}$$

The first uses the steady state $\bar{v}_{pre} = (1 - \bar{\delta})[(1 - \bar{q})\bar{v} + s(1 - \bar{u})]$, and the third uses $\partial q/\partial \theta = -\eta_L \bar{q}/\bar{\theta}$. Hence, with $\delta_{t-1} - \bar{\delta} = \bar{\delta}\tilde{\delta}_{t-1}$,

$$v_{pre,t} - \bar{v}_{pre} = -\frac{\bar{v}_{pre}}{1 - \bar{\delta}} \bar{\delta} \tilde{\delta}_{t-1} + (1 - \bar{\delta})[(1 - \bar{q})\bar{v} \hat{v}_{t-1} + \eta_L \bar{q} \bar{v} \hat{\theta}_{t-1} - s\bar{u} \hat{u}_{t-1}].$$

Dividing by $\bar{v}_{pre} = \bar{v} - \bar{e}$,

$$\hat{v}_{pre,t} = -\frac{\bar{\delta}}{1 - \bar{\delta}} \tilde{\delta}_{t-1} + \frac{1 - \bar{\delta}}{\bar{v} - \bar{e}} \left[(1 - \bar{q})\bar{v} \hat{v}_{t-1} + \eta_L \bar{q} \bar{v} \hat{\theta}_{t-1} - s\bar{u} \hat{u}_{t-1} \right] \quad (18)$$

Entry replaces the positions destroyed, so its weight in (17), $\bar{\delta}(\bar{v} + 1 - \bar{u})/\bar{v}$, is proportional to $\bar{\delta}$: 0.073 at D1 and 0.831 at $\bar{\delta} = \bar{\tau}$. The predetermined stock is the rest, 0.927 and 0.169. In (18), a one-percent rise in δ_{t-1} destroys about $\bar{\delta}$ percent of the predetermined stock, while entrants posted at t are not exposed. The bracket collects the changes in the sources of the predetermined stock, in units of vacancies. More vacancies at $t - 1$ leave more unfilled ones $((1 - \bar{q})\bar{v} \hat{v}_{t-1})$. A tighter market fills fewer of them $(\eta_L \bar{q} \bar{v} \hat{\theta}_{t-1})$; after a destruction shock $\hat{\theta} < 0$, so vacancies fill faster and fewer carry over. And reposts come from the employed, who are fewer when u is higher $(-s\bar{u} \hat{u}_{t-1})$.

At free entry. With $\hat{v}_{t-1} = \hat{\theta}_{t-1} + \hat{u}_{t-1}$, the bracket in (18) becomes $(1 - \bar{q} + \eta_L \bar{q})\bar{v} \hat{\theta}_{t-1} + [(1 - \bar{q})\bar{v} - s\bar{u}]\hat{u}_{t-1}$. Equation (17) then determines entry, not the vacancy stock:

$$\hat{e}_t = \frac{\bar{v}}{\bar{\delta}(\bar{v} + 1 - \bar{u})} \left\{ \hat{v}_t - \left[1 - \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}} \right] \hat{v}_{pre,t} \right\}, \quad \hat{v}_t = \hat{\theta}_t + \hat{u}_t.$$

When the entrant weight is small, as at D1, a given gap between the target stock and the predetermined stock requires a large percentage change in entry.

4.4. Tightness

Taking logs of (8),

$$\boxed{\hat{v}_t = \hat{\theta}_t + \hat{u}_t} \quad (19)$$

4.5. Weights at the calibration

The numbers use our calibration, not GS's. It holds $\bar{f}$, $\bar{q}$ and $\bar{\tau}$ fixed and recalibrates s and ϕ at each $\bar{\delta}$. The discount rate is $r = 0.0033$ monthly.

	D1 ($\bar{\delta} = 3.2\%/yr$)	CK-like ($\bar{\delta} = \bar{\tau}$)
$\bar{\delta}/(r + \bar{\delta})$ (flow value)	0.453	0.905
$\bar{f}\bar{\delta}/\bar{\tau}$ (unemployment)	0.036	0.423
$(1 - s)\bar{\delta}/\bar{\tau}$ (separations)	0.085	1.000
$(1 - \bar{\delta})\bar{f}$	0.410	0.410
λ	0.559	0.559
ω	0.164	0.164
$1 - \bar{\delta}(\bar{v} + 1 - \bar{u})/\bar{v}$ (predetermined)	0.927	0.169
$\bar{\delta}(\bar{v} + 1 - \bar{u})/\bar{v}$ (entrants)	0.073	0.831

Table 1: Weights of the log-linear system, per unit of $\tilde{\delta}$.

Moving from $\bar{\delta} = \bar{\tau}$ to D1 cuts the unemployment loading by a factor of 12 but the flow-value loading only by a factor of 2. The first is proportional to $\bar{\delta}$. The second saturates, because $\bar{\delta}/(r + \bar{\delta})$ falls in proportion to $\bar{\delta}$ only once $\bar{\delta}$ is below r . The entrant share falls by the same factor of 11 as the destruction share of separations, since $\bar{e}/\bar{v} = (\bar{\delta}/\bar{\tau})[\bar{\tau} + (1 - \bar{\delta})\bar{q}]$ with a bracket fixed by our targets. So the unemployment loading and the entrant share both scale with $\bar{\delta}/\bar{\tau}$, while the flow-value loading does not. The other weights are essentially the same in both economies.

5. Proposition

With λ and ω as defined in Section 4, define

$$a \equiv - \frac{(1+r)\frac{\bar{\delta}}{1-\bar{\delta}} + \frac{\bar{\delta}}{r+\bar{\delta}} \left[1+r-\rho(1-\bar{\tau}) + \rho\phi(1-\bar{\delta})\bar{f} + \rho(1-\phi)(1-\bar{\delta})\bar{q} \right]}{\eta_L \left[1+r-\rho(1-\bar{\tau}) \right] + \rho\phi(1-\bar{\delta})\bar{f}} \quad (20)$$

Lemma 1 (Tightness under infinitely elastic vacancy creation). *Assume the setting of Section 1 with $\xi \rightarrow \infty$, and suppose $\phi(1 - \bar{\delta})\bar{f} < \eta_L(2 + r - \bar{\tau})$, which holds whenever $\phi \leq \eta_L$. Then the log-linear system (14)–(19) has a unique bounded solution, in which tightness depends only on the destruction shock:*

$$\hat{\theta}_t = a \tilde{\delta}_t, \quad a < 0.$$

The endogenous states u_t and $v_{pre,t}$ do not enter the policy rule for tightness.

Proof. A general linear policy rule would take the form $\hat{\theta}_t = a \tilde{\delta}_t + b_u \hat{u}_t + b_v \hat{v}_{pre,t}$. Equation (14) contains neither u nor v_{pre} , because free entry pins Q and $K_t = x_m(r + \delta_t)/(1 + r)$ depends only on the exogenous δ_t . Dividing it by η_L ,

$$\begin{aligned} \hat{\theta}_t &= \Phi \mathbb{E}_t \hat{\theta}_{t+1} + c_0 \tilde{\delta}_t + c_1 \mathbb{E}_t \tilde{\delta}_{t+1}, & \Phi &\equiv \frac{1}{1+r} \left[1 - \bar{\tau} - \frac{\phi(1 - \bar{\delta})\bar{f}}{\eta_L} \right], \\ c_0 &\equiv -\frac{1}{\eta_L} \left[\frac{\bar{\delta}}{r + \bar{\delta}} + \frac{\bar{\delta}}{1 - \bar{\delta}} \right], & c_1 &\equiv \frac{1 - \bar{\tau} - \phi(1 - \bar{\delta})\bar{f} - (1 - \phi)(1 - \bar{\delta})\bar{q}}{\eta_L(1 + r)} \frac{\bar{\delta}}{r + \bar{\delta}}. \end{aligned}$$

Iterating forward T periods,

$$\hat{\theta}_t = \sum_{k=0}^{T-1} \Phi^k \mathbb{E}_t (c_0 \tilde{\delta}_{t+k} + c_1 \tilde{\delta}_{t+k+1}) + \Phi^T \mathbb{E}_t \hat{\theta}_{t+T}.$$

Here $\Phi < (1 - \bar{\tau})/(1 + r) < 1$. And $\Phi > -1$ if and only if $1 - \bar{\tau} - \phi(1 - \bar{\delta})\bar{f}/\eta_L > -(1 + r)$, that is, $\phi(1 - \bar{\delta})\bar{f} < \eta_L(2 + r - \bar{\tau})$, which is the assumption. It holds whenever $\phi \leq \eta_L$, because $(1 - \bar{\delta})\bar{f} < 1 < 2 + r - \bar{\tau}$. For a bounded solution the last term vanishes as $T \rightarrow \infty$, so the unique bounded solution is the forward sum. It depends only on current and expected future $\tilde{\delta}$, so $b_u = b_v = 0$. With $\mathbb{E}_t \tilde{\delta}_{t+k} = \rho^k \tilde{\delta}_t$, every term of the sum is proportional to $\tilde{\delta}_t$, so $\hat{\theta}_t = a \tilde{\delta}_t$ for a constant a .

To find a , substitute $\hat{\theta}_t = a \tilde{\delta}_t$, $\mathbb{E}_t \hat{\theta}_{t+1} = a \rho \tilde{\delta}_t$ and $\mathbb{E}_t \tilde{\delta}_{t+1} = \rho \tilde{\delta}_t$ into (14), divide by $\tilde{\delta}_t$ and multiply by $1 + r$:

$$\begin{aligned} (1 + r) \eta_L a &= -(1 + r) \left[\frac{\bar{\delta}}{r + \bar{\delta}} + \frac{\bar{\delta}}{1 - \bar{\delta}} \right] + \rho \left[1 - \bar{\tau} - \phi(1 - \bar{\delta})\bar{f} - (1 - \phi)(1 - \bar{\delta})\bar{q} \right] \frac{\bar{\delta}}{r + \bar{\delta}} \\ &\quad + \rho \left[(1 - \bar{\tau})\eta_L - \phi(1 - \bar{\delta})\bar{f} \right] a. \end{aligned}$$

Collecting the terms in a on the left,

$$a \left\{ \eta_L [1 + r - \rho(1 - \bar{\tau})] + \rho \phi(1 - \bar{\delta})\bar{f} \right\} = -(1 + r) \frac{\bar{\delta}}{1 - \bar{\delta}}$$

$$- \frac{\bar{\delta}}{r + \bar{\delta}} \left[1 + r - \rho(1 - \bar{\tau}) + \rho\phi(1 - \bar{\delta})\bar{f} + \rho(1 - \phi)(1 - \bar{\delta})\bar{q} \right],$$

which is (20). Since $\rho < 1$, $1 + r - \rho(1 - \bar{\tau}) > 0$. Every term of the numerator and denominator of (20) is then positive, so $a < 0$. $\square$

Reading (20). The term common to numerator and denominator,

$$1 + r - \rho(1 - \bar{\tau}) = r + \bar{\tau} + (1 - \rho)(1 - \bar{\tau}),$$

is the rate at which a match discounts the shock's effects on its future payoffs: the discount rate, plus the separation rate, plus the rate at which the shock dies out. In the denominator it multiplies η_L , the response of the expected cost of a hire to tightness. The second term, $\rho\phi(1 - \bar{\delta})\bar{f}$, is the wage feedback: lower tightness lowers the worker's effective job-finding rate and hence her outside option, which props up the firm's surplus. At $\rho = 1$ the denominator is $\eta_L(r + \bar{\tau}) + \phi(1 - \bar{\delta})\bar{f}$, the same combination of $r + \bar{\tau}$ and $\phi(1 - \bar{\delta})\bar{f}$ that appears in the steady-state job creation condition once the wage is substituted.

The numerator has a survival part, $(1 + r)\bar{\delta}/(1 - \bar{\delta})$, which is small, and a flow-value part, $\bar{\delta}/(r + \bar{\delta})$, times a bracket. The bracket starts from the same effective rate. This already nets out that a persistently high flow value raises the hiring cost a continuing match saves, through $\rho(1 - \bar{\tau})$. Two persistence terms then add to it: part of that saving goes to the worker, $\rho\phi(1 - \bar{\delta})\bar{f}$, and a matched firm forgoes the higher flow value, $\rho(1 - \phi)(1 - \bar{\delta})\bar{q}$. At $\rho = 0$ the bracket and the denominator's effective rate are both $1 + r$, and $a = -[\bar{\delta}/(r + \bar{\delta}) + \bar{\delta}/(1 - \bar{\delta})]/\eta_L$.

Remark 1 (Block recursivity). *Given Lemma 1, the states are solved backward. Equation (16) gives $\hat{u}$, and (19) gives $\hat{v} = \hat{\theta} + \hat{u}$. The predetermined stock $v_{pre,t} = (1 - \delta_{t-1})\{[1 - q(\theta_{t-1})]\theta_{t-1}u_{t-1} + s(1 - u_{t-1})\}$ depends only on $(u_{t-1}, \theta_{t-1}, \delta_{t-1})$ and has no own lag. It feeds into nothing except entry, $e_t = v_t - v_{pre,t}$. The system thus has one unstable root, $1/\Phi$, for one jump variable, and one stable root, λ , for one dynamic state, u , so the Blanchard–Kahn condition holds.*

*The structure breaks with finite ξ . Entry $e = G(Q)$ then links v_{pre} to Q , and Q enters the job creation condition. Tightness depends on the states, the policy rule must include state coefficients, and the closed form is lost. The same happens when the conditions of **prop:independence** fail, through N , C or χ^c .*

With a from Lemma 1, define the ratio of the unemployment loading to the tightness loading,

$$R \equiv \frac{\bar{f} \bar{\delta} / \bar{\tau}}{|a|}.$$

In the CK case, $\bar{\delta} = \bar{\tau}$ and $R = \bar{f}/|a|$.

Proposition 1 (Destruction shocks under infinitely elastic vacancy creation). *Under the assumptions of Lemma 1, consider an innovation $\varepsilon_0 > 0$. Let $\lambda \equiv 1 - (1 - \bar{\delta})\bar{f} - \bar{\tau} \in [0, 1)$ be the persistence of unemployment, one minus the effective job-finding rate and the separation rate, and let $\omega \equiv (1 - \eta_L)(1 - \bar{\delta})\bar{f}$ be the effective job-finding rate times its elasticity to tightness. To first order,*

$$\hat{\theta}_t = a \rho^t \varepsilon_0, \quad \hat{u}_t = |a|(\omega + R) \varepsilon_0 \sum_{j=0}^{t-1} \lambda^{t-1-j} \rho^j, \quad \hat{v}_t = \hat{\theta}_t + \hat{u}_t,$$

so unemployment is unchanged on impact and rises thereafter, while vacancies fall on impact. Moreover:

1. Vacancies fall at $h = 1$ if and only if

$$\rho > \omega + R. \tag{21}$$

Once vacancies rise above steady state, they stay above it.

2. The slope of $v_t - \bar{v}$ on $u_t - \bar{u}$ along the impulse response is

$$\bar{\theta} \left[1 - \frac{\rho(1 - \lambda^2)}{(1 + \rho\lambda)(\omega + R)} \right], \tag{22}$$

which is negative if and only if

$$\frac{\rho(1 - \lambda^2)}{1 + \rho\lambda} > \omega + R. \tag{23}$$

In particular, both conditions fail when $\rho = 0$: an iid shock lowers vacancies only on impact, after which $v_t = \bar{\theta}u_t$ and the slope equals $\bar{\theta} > 0$.

Both conditions compare the persistence of the shock with the same threshold, $\omega + R$. At $h = 1$, the tightness fall that survives one period, $|a|\rho$, must exceed the rise in unemployment. That rise has two parts: the hires lost at date 0, $\omega|a|$, and the destruction itself, $\bar{f}\bar{\delta}/\bar{\tau} = R|a|$. So $\rho > \omega$ is necessary, and R is the extra persistence needed to outweigh destruction. Over the whole path, persistence is discounted by unemployment's own persistence, since $\rho(1 - \lambda^2)/(1 + \rho\lambda) < \rho$, so (23) is the stricter condition.

Proof. The innovation sets $\tilde{\delta}_t = \rho^t \varepsilon_0$, so by Lemma 1 $\hat{\theta}_t = a\rho^t \varepsilon_0$.

Step 1: unemployment. u_0 is predetermined, so $\hat{u}_0 = 0$. Substituting $\hat{\theta}_{t-1} = -|a|\rho^{t-1}\varepsilon_0$ and $\tilde{\delta}_{t-1} = \rho^{t-1}\varepsilon_0$ into (16), and using $\bar{f}\bar{\delta}/\bar{\tau} = R|a|$,

$$\hat{u}_t = \lambda\hat{u}_{t-1} + \bar{f}\frac{\bar{\delta}}{\bar{\tau}}\rho^{t-1}\varepsilon_0 + \omega|a|\rho^{t-1}\varepsilon_0 = \lambda\hat{u}_{t-1} + |a|(\omega + R)\rho^{t-1}\varepsilon_0.$$

Iterating from $\hat{u}_0 = 0$ gives $\hat{u}_1 = |a|(\omega + R)\varepsilon_0$, $\hat{u}_2 = |a|(\omega + R)(\lambda + \rho)\varepsilon_0$, and in general $\hat{u}_t = |a|(\omega + R)\varepsilon_0 g_t$ with $g_t \equiv \sum_{j=0}^{t-1} \lambda^{t-1-j}\rho^j$. Since $\lambda, \rho \geq 0$ and $g_1 = 1$, $\hat{u}_t > 0$ for $t \geq 1$. By (19), $\hat{v}_t = \hat{\theta}_t + \hat{u}_t$, and $\hat{v}_0 = a\varepsilon_0 < 0$.

Step 2: part 1. With $g_1 = 1$,

$$\hat{v}_1 = -|a|\rho\varepsilon_0 + |a|(\omega + R)\varepsilon_0 = |a|(\omega + R - \rho)\varepsilon_0,$$

which is negative if and only if (21) holds. For $t \geq 1$ and $\rho > 0$, divide $\hat{v}_t = |a|[-\rho^t + (\omega + R)g_t]\varepsilon_0$ by $|a|\rho^t\varepsilon_0 > 0$, and use $g_t/\rho^t = \rho^{-1} \sum_{i=0}^{t-1} (\lambda/\rho)^i$:

$$\hat{v}_t < 0 \iff \rho > (\omega + R) \sum_{i=0}^{t-1} \left(\frac{\lambda}{\rho}\right)^i. \quad (24)$$

The right side is strictly increasing in t . So once the inequality fails, it fails at all later dates. For $\rho = 0$, $\hat{v}_t = \hat{u}_t > 0$ for all $t \geq 1$.

Step 3: part 2. In levels, $u_t - \bar{u} = \bar{u}|a|(\omega + R)\varepsilon_0 g_t$ and $v_t - \bar{v} = \bar{v}|a|[-\rho^t + (\omega + R)g_t]\varepsilon_0$, with $g_0 = 0$. For $\rho \neq \lambda$, $g_t = (\rho^t - \lambda^t)/(\rho - \lambda)$; the case $\rho = \lambda$ follows by continuity. Summing geometric series,

$$\sum_{t \geq 0} \rho^t g_t = \frac{\rho}{(1 - \rho^2)(1 - \rho\lambda)}, \quad \sum_{t \geq 0} g_t^2 = \frac{1 + \rho\lambda}{(1 - \rho^2)(1 - \lambda^2)(1 - \rho\lambda)}.$$

The first uses $\frac{1}{1 - \rho^2} - \frac{1}{1 - \rho\lambda} = \frac{\rho(\rho - \lambda)}{(1 - \rho^2)(1 - \rho\lambda)}$. The second uses $(1 - \rho\lambda)(1 - \lambda^2) - 2(1 - \rho^2)(1 - \lambda^2) + (1 - \rho^2)(1 - \rho\lambda) = (\rho - \lambda)^2(1 + \rho\lambda)$. Therefore

$$\frac{\sum_t (v_t - \bar{v})(u_t - \bar{u})}{\sum_t (u_t - \bar{u})^2} = \frac{\bar{v}}{\bar{u}} \left[1 - \frac{1}{\omega + R} \frac{\sum_t \rho^t g_t}{\sum_t g_t^2} \right] = \bar{\theta} \left[1 - \frac{\rho(1 - \lambda^2)}{(1 + \rho\lambda)(\omega + R)} \right],$$

which is (22). It is negative if and only if (23) holds.

Step 4: $\rho = 0$. The left sides of (21) and (23) are zero, while $\omega + R > 0$, so both conditions fail. Since $\hat{\theta}_t = a \cdot 0^t \varepsilon_0 = 0$ for $t \geq 1$, $\hat{v}_t = \hat{u}_t$ and $v_t = \bar{\theta}u_t$ to first order. The slope (22) is then $\bar{\theta}$. $\square$

Corollary 1 (Vacancies below steady state at every horizon). *Under the assumptions of Proposition 1, $\hat{v}_t < 0$ for every $t \geq 0$ if and only if $\rho - \lambda \geq \omega + R$. This requires $\rho > \lambda + \omega$, which is 0.72 at our calibration.*

Proof. In (24), if $\rho \leq \lambda$ the right side grows without bound, so vacancies eventually rise above steady state. If $\rho > \lambda$ it converges from below to $(\omega + R)\rho/(\rho - \lambda)$, so the inequality holds at every t if and only if $\rho - \lambda \geq \omega + R$. $\square$

6. Interpretation

Why tightness falls. At $\rho = 0$, (14) reads $\eta_L \hat{\theta}_0 = -[\bar{\delta}/(r + \bar{\delta}) + \bar{\delta}/(1 - \bar{\delta})]\tilde{\delta}_0$. The flow value of a vacancy, and hence the cost of a hire, rises by $\bar{\delta}/(r + \bar{\delta})$ percent per percent of δ , which is 0.45 at D1. The value of a hire falls only through survival, by the small amount $\bar{\delta}/(1 - \bar{\delta})$. For the two to stay equal, the expected time to fill, $1/q$, must fall, so tightness falls.

The iid case. Tightness and vacancies fall only at date 0, while unemployment is predetermined. At date 1 the flow value is back to $\bar{K}$ and $\theta_1 = \bar{\theta}$. Unemployment rises, and $v_1 = \bar{\theta}u_1$ rises with it as entry refills the stock. In the (u, v) plane the economy drops straight down, jumps up and to the right, and returns along the ray $v = \bar{\theta}u$. At D1, of the 0.16% rise in $\hat{u}_1$ per 1% shock, 0.04 comes from destruction, $(\bar{f}\bar{\delta}/\bar{\tau})\tilde{\delta}_0$, and 0.12 from the date-0 collapse in hiring, $\omega|\hat{\theta}_0|$.

The persistent case. Tightness stays low while destruction stays high, so vacancies stay below steady state while unemployment builds up. Once tightness has recovered enough, $v = \theta u$ turns positive and stays positive.

Where $\bar{\delta}$ relative to $\bar{\tau}$ enters. The destruction rate enters both conditions only through R , the ratio of the unemployment loading, $\bar{f}\bar{\delta}/\bar{\tau}$, to the tightness loading, $|a|$. The second is roughly proportional to $\bar{\delta}/(r + \bar{\delta})$, so R is roughly proportional to $(r + \bar{\delta})/\bar{\tau}$. Lowering $\bar{\delta}$ relative to $\bar{\tau}$ shrinks the unemployment loading in proportion, while the tightness loading shrinks much less (Table 1), so R falls. At D1, $(r + \bar{\delta})/\bar{\tau} = 0.19$ and $R = 0.045$, so the threshold $\omega + R$ is barely above $\omega = 0.164$. With $\bar{\delta} = \bar{\tau}$ (CK, $s = 0$), $(r + \bar{\delta})/\bar{\tau} = 1.11$ and $R = \bar{f}/|a| = 0.289$: the vacancy dies with the match, so its flow value cannot respond much more than unemployment does, and the threshold rises to 0.453.

7. Numbers

The calibration uses $\rho = 0.592$, and responses are per 1% shock ($\tilde{\delta}_0 = 0.01$), in percent. At D1, $a = -0.790$ (iid: -0.759). At $\bar{\delta} = \bar{\tau}$, $a = -1.464$ (iid: -1.561).

t	D1			CK-like ($\bar{\delta} = \bar{\tau}$)		
	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$
0	-0.79	0.00	-0.79	-1.46	0.00	-1.46
1	-0.47	0.17	-0.30	-0.87	0.66	-0.20
2	-0.28	0.19	-0.09	-0.51	0.76	0.25
3	-0.16	0.16	0.00	-0.30	0.66	0.36
4	-0.10	0.13	0.03	-0.18	0.51	0.33

Table 2: Impulse responses to a persistent destruction shock ($\rho = 0.592$).

	ω	R	$\omega + R$	$\rho(1 - \lambda^2)/(1 + \rho\lambda)$	slope	ρ needed for slope < 0
D1	0.164	0.045	0.209	0.306	-0.24	$\rho > 0.37$
CK-like	0.164	0.289	0.453	0.306	0.17	none

Table 3: Conditions of Proposition 1 at $\rho = 0.592$. The slope after an iid shock is $\bar{\theta} = 0.51$ in both economies.

Condition (21), $\rho > \omega + R$, holds in both economies, so the sign at $h = 1$ does not separate them. The path condition (23) does: 0.306 exceeds the threshold at D1 (0.209) but not at $\bar{\delta} = \bar{\tau}$ (0.453).

8. Remarks

Remark 2 (Holding K fixed). *If the vacancy cost were a constant flow, rather than the flow value of an asset that δ_t depreciates, the $\bar{\delta}/(r + \bar{\delta})$ terms would drop out of (14) and (20). Only the survival effect would remain, and $|a| = (1 + r)[\bar{\delta}/(1 - \bar{\delta})]/\{\eta_L[1 + r - \rho(1 - \bar{\tau})] + \rho\phi(1 - \bar{\delta})\bar{f}\}$ is small. At our calibration the path slope is then positive in every economy (about 0.49 at $\rho = 0.592$). This is the flow-cost DMP case.*

Remark 3. *A permanent shock ($\rho \rightarrow 1$) recovers the steady-state comparison. It hides the last statement of Proposition 1: without persistence, the response is positively sloped after impact.*

Remark 4. *prop:ds_asymmetry* Part 3 in the draft gives “ $\bar{\delta}_e/\bar{\tau}$ small” as a sufficient condition. In the iid free-entry case, entry refills the stock at $t + 1$, and that condition fails for every $\bar{\delta}_e/\bar{\tau}$.